

Qingkai Dong

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Professional Summary

Statistics Ph.D. student, expected to graduate in May 2027, with Servier Pharmaceuticals experience in individual patient data reconstruction and deep learning methods for interim analysis. Develops rare-feature subsampling methods, simulation studies, and reproducible software in R and Python.

Education

University of Connecticut Ph.D. in Statistics Advisors: Prof. HaiYing Wang and Prof. Jun Yan	Aug 2023 – Expected May 2027 Storrs, CT, USA
Zhongnan University of Economics and Law MS in Mathematical Statistics	Sep 2020 – May 2023 Wuhan, China
Qingdao University BS in Applied Statistics	Sep 2016 – May 2020 Qingdao, China

Technical Skills

Statistical Methods: survival analysis; interim analysis & conditional power; design simulation; big-data subsampling; model averaging; variable selection

Programming: R (package development), Python (machine learning / deep learning), SAS, Git, LaTeX

Industry and Research Experience

Global Biometrics Intern Servier Pharmaceuticals	Jun 2026 – Aug 2026
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- Built a modular workflow to reconstruct synthetic individual patient data from published Kaplan–Meier curves and summary statistics using target-matching loss functions.
- Demonstrated close agreement with reported survival metrics and evaluated accuracy for additional survival quantities in simulation studies; applied the workflow to internal data.

Research Assistant (Academia–Industry Collaboration) Servier Pharmaceuticals & University of Connecticut	Mar 2024 – Dec 2025
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- Developed and implemented a covariate-informed deep learning framework to predict event times for censored participants and augment interim survival data for conditional-power estimation and futility decisions.
- Designed evaluation metrics for conditional-power forecast accuracy and futility decision quality. Conducted simulation studies to characterize when performance improves or degrades under varied censoring, covariate informativeness/type, and prediction range.
- Co-first-authored the methodology manuscript, under revision at *Statistics in Medicine*.

Research Assistant (Statistical Methodology) University of Connecticut	Aug 2023 – Present
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- Developed rare-feature-aware subsampling methods for regression to preserve representation of rare binary covariates. Implemented the methods in the subsampling R package.
- Established theory linking the efficiency of rare-covariate coefficient estimation to the number of rare observations, explaining numerical instability and slow convergence.
- Evaluated the methods through simulation studies and applications to rare disease data. First-authored a methodology manuscript (under review at *JASA*).

Research Assistant (Breast Cancer Survivorship) Departments of Statistics and Human Development and Family Sciences, University of Connecticut	Jul 2024 – Aug 2024
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- Retrieved, cleaned, and integrated All of Us EHR, survey, and physical-measurement data to prepare inputs for subsequent statistical modeling.
- Contributed to study framing on social determinants of health, frailty, and accelerated aging among breast cancer survivors.

Cross-Disciplinary Collaboration (Chemical Sensor Data Analytics) Departments of Statistics and Chemistry, University of Connecticut	Mar 2024 – May 2024
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- Applied clustering and statistical analysis to porphyrinoid-sensor fluorescence data to assess compound differentiation and inform sensor selection.
- Produced interpretable summaries and R visualizations to support collaborators' experimental decisions.

Cross-Disciplinary Collaboration (Large Language Model Explainability) 2025 – Present
 University of Connecticut & collaborators

- Curated benchmarks and released an accompanying GitHub repository for a survey of time series explainability and LLM-enabled semantic explanations.
- Coauthored the survey manuscript, currently under review at *ACM Computing Surveys*.

Software

R package subsampling (CRAN)

- Implemented sampling-probability computation, pilot sampling, and two-step designs for GLMs, quantile regression, softmax regression, and rare-event/rare-feature settings.
- Delivered a modular R API with documentation, vignettes, and reproducible examples.

Selected Papers

Dong Q, Yan J, Zhang Q, Wang H. Rare-feature-adaptive subsampling for big data regression. Under review at *JASA*.
 Wang D[†], **Dong Q**[†], Zhang Q, Hu Q, Wang H. Random Subsampling: Inferential Targets, Estimators, and Sampling Procedures. *Under review*. [†]Equal contribution.
Dong Q[†], Man A[†], Hua Z, Teng Z, Wang H, Yan J. Predictive-Modeling-Assisted Interim Decision-Making in Adaptive Trials with Censored Survival Outcomes. Under revision at *Statistics in Medicine*. [†]Equal contribution.
Dong Q, Liu B, Zhao H. Weighted Least Squares Model Averaging for Accelerated Failure Time Models. *Computational Statistics and Data Analysis*, 2023.
 Zhao H, **Dong Q**. Variable Selection for Additive Hazards Model with Current Status Data. *Journal of Systems Science and Mathematical Sciences*, 2022.
 Zhao H, Liu B, **Dong Q**. Jackknife Model Averaging of AFT Model with Current Status Data. *Acta Mathematicae Applicatae Sinica*, 2023.
 Chen Z, Lucchesi G, **Dong Q**, Zheng X, Song D, Wen Q, Cheng W, Ni J, Luo D. From Signals to Semantics: A Survey on Time Series Explainability through a Human-Cognitive Lens. Manuscript under review at *ACM Computing Surveys*, 2025.

Teaching Experience

University of Connecticut

Principal Instructor (Upcoming Appointment) Spring 2027
 Scheduled to teach STAT 3255: Introduction to Data Science.

Teaching Assistant

Sep 2026 – Dec 2026
 Supporting STAT 1100Q: Elementary Concepts of Statistics and STAT 3445: Introduction to Mathematical Statistics II.

Principal Instructor

Jan 2026 – May 2026
 STAT 3675Q: Statistical Computing.
 Taught R/RStudio for data management, visualization, reproducible analysis, and statistical modeling.

Academic Presentations

Invited talks (2026): Joint Statistical Meetings (JSM); New England Statistics Symposium (NESS); ICSA Applied Statistics Symposium; SILVER Online Lab Meeting, Department of Epidemiology & Public Health, University of Maryland School of Medicine.
Posters: New England Rare Disease Statistics Workshop (NERDS, 2025, Oct 2026); NESS (2026); The Design and Analysis of Experiments (DAE, 2026); Dahshu Data Science Symposium (2025).

Academic Service

Journal Reviewer: *Computational Statistics and Data Analysis*; *Journal of Computational Science*; *Statistical Papers*; *Sankhya B*; *Journal of Systems Science and Mathematical Sciences (Chinese)*

Awards

UConn Graduate School Conference Participation Award Jul 2026
New England Statistics Symposium (NESS) Poster Award 2026
ICSA Applied Statistics Symposium Student Paper Award 2026
Predoc Fellowship - University of Connecticut 2025
First-class Scholarship - Zhongnan University of Economics and Law 2020, 2022